

ATRI SUKUL

Haldia, West Bengal 721635

+91 8900519836 | atrisukul1508@gmail.com

linkedin.com/in/atrisukul | github.com/AtriSukul1508 | scholar.google.com/atrisukul | atrisukul1508.github.io

SUMMARY

Machine Learning Practitioner working on healthcare AI at IIT Madras, where I am building clinical NLP systems to make medical information more accessible and useful. I enjoy understanding ideas from research papers and putting them into practice, and have co-authored research published in IEEE TGRS and the Proceedings of the CVPR Conference.

EDUCATION

Indian Institute of Technology, Tirupati <i>M.Tech in Data Science and Artificial Intelligence</i> CGPA: 8.94	07/2025 – Present Tirupati, India
Jalpaiguri Government Engineering College <i>B.Tech in Computer Science and Engineering</i> CGPA: 9.33	08/2021 – 06/2025 Jalpaiguri, India
Mahishadal Raj College <i>B.Sc in Mathematics</i> CGPA: 8.44	07/2018 – 08/2021 Mahishadal, India

EXPERIENCE

Project Associate IC&SR, IIT Madras (with UHG Optum) <i>Healthcare AI / NLP</i>	08/2025 – Present Chennai, India (Remote)
- Building an NLP pipeline that transforms unstructured EHR discharge reports into knowledge graphs, making relevant clinical information easier to identify and use. - Designing a context-grounded GraphRAG question-answering system for clinical records to support faster and more reliable information retrieval.	
Student Fellow Indian National Academy of Engineering (INAE) <i>Mentoring of Engineering Students by INAE Fellow</i>	02/2024 – 03/2024 India
- Selected under the “Mentoring of Engineering Students by INAE Fellow-2021” scheme, one of only 60 students chosen across India that year for the fellowship. - Mentored by Prof. Umapada Pal, ISI Kolkata, under the scheme (2023–24); built a frequency-domain generative framework for hyperspectral image classification, reaching 95.52% (Pavia) and 99.05% (Salinas) and surpassing models such as SpectralDiff.	

SKILLS

Technologies: PyTorch, LangChain, FastAPI, OpenCV, Pandas, NumPy, Scikit-Learn, ReactJS, NodeJS

Languages: Python, C, C++, JavaScript, SQL

Databases: MySQL, MongoDB

Specializations: LLM, NLP, Computer Vision, Deep Learning, Remote Sensing

PROJECTS

clawss package FastAPI, React, PyTorch, Ollama/OpenRouter GitHub	05/2026
A browser-based modular notebook where cells behave like Python files with isolated namespaces and cross-cell imports; adds a cell-dependency graph, PyTorch architecture visualization, and notebook-aware AI generation via local or API-based models.	
Cross Hyperspectral and LiDAR Attention Transformer PyTorch GitHub	08/2023
Implemented the full codebase for the IEEE TGRS paper, realizing the proposed extended self-attention method for land-use and land-cover classification.	

PUBLICATIONS

- S. K. Roy, **Atri Sukul**, A. Jamali, J. Haut, P. Ghamisi. “Cross Hyperspectral and LiDAR Attention Transformer: An Extended Self-Attention for Land Use and Land Cover Classification.” *IEEE Transactions on Geoscience & Remote Sensing*, 2024. [IEEE Xplore]
- S. Mitra, **Atri Sukul**, S. K. Roy, P. Singh, V. K. Verma. “ScoreCAM++: Gated Score-Weighted Visual Explanations for CNNs.” *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2700-2709. [CVPR Open Access]

ACHIEVEMENTS

- Smart India Hackathon (SIH) 2022 Winner** | Team Odyssey (PS - Virtual Excavation Game). [Link]
- Hack4Bengal 2.0** | Secured the 1st Runner-Up spot for the Flow Track in Hack4Bengal 2.0 [Link]

LEADERSHIP AND VOLUNTEERING

- AI/ML Lead**, Google Developer Student Club, JGEC (2023–2024).
- Deputy Secretary**, National Service Scheme (NSS) Unit, JGEC.
- Lead Organizer**, JWOC 2024 | a nationwide open-source program with 1900+ participants.